

NarrowGap: Reducing Bottlenecks for End-to-End Entanglement Distribution in Quantum Networks

Zhonghui Li¹, Member, IEEE, Jian Li², Senior Member, IEEE, Kaiping Xue¹, Senior Member, IEEE,
Lutong Chen¹, Graduate Student Member, IEEE, Nenghai Yu¹,
Qibin Sun¹, Fellow, IEEE, and Jun Lu¹

Abstract—Quantum networks, which work by establishing entanglement between distant quantum end nodes (known as end-to-end entanglement distribution), are the promising infrastructure for quantum applications. However, the inherent loss in quantum channels and quantum decoherence contribute to the scarcity of entanglement resources in quantum networks. Consequently, there is an inevitable gap between available entanglement resources and requests' demands, significantly hindering concurrent end-to-end entanglement distributions. In this paper, we present NarrowGap, an end-to-end entanglement distribution design that can alleviate the negative impact of entanglement resource scarcity on the request service capability of quantum networks. At the heart of NarrowGap, the resource transfer scheme (RTS) is designed to transfer idle entanglement resources to boost the bottlenecks' capacities based on the unique feature of entanglement swapping, thus narrowing the gap between available entanglement resources and requests' demands for end-to-end entanglements. Besides, NarrowGap presents a resource allocation scheme (RAS) to guarantee fairness, considering both the success probability of end-to-end entanglement distribution and each request's demand, to address resource competition in bottlenecks. Extensive simulations demonstrate that NarrowGap outperforms three representative schemes and can achieve more than twice the performance improvement in request service rate.

Index Terms—Quantum networks, entanglement distribution, entanglement swapping, congestion mitigation.

I. INTRODUCTION

THE dramatic development of quantum information technology promotes the emergence of quantum networks [1], [2]. In terms of physical structure, a quantum network is the combination of numerous quantum nodes capable of preparing, transmitting, storing, and processing quantum bits

(i.e., *qubits*). These nodes are interconnected via quantum channels in synergy with classical ones to act as a promising platform for various quantum applications, such as quantum secure communication [3], quantum computing [4]. Notably, most quantum applications require distant quantum end nodes to be entangled by sharing entangled qubit pairs (hereafter referred to as entangled pairs). Hence, establishing entanglement between distant quantum end nodes, i.e., *end-to-end entanglement distribution*, becomes one of the essential building blocks of quantum networks [5].

Quantum repeaters [6] play a pivotal role in end-to-end entanglement distribution. Although directly distributing entangled pairs between adjacent quantum nodes (i.e., link-level entanglement distribution) has been experimentally demonstrated in recent years, the photon loss of quantum channels significantly limits entanglement distribution over long distances. Hence, quantum repeaters are required in quantum networks. Different from signal amplification or generation in classical communications, quantum repeaters work by the innovative technique known as entanglement swapping [7]. Entanglement swapping is essentially a joint measurement operation with a probability feature, which can generate entanglement between two non-entangled nodes without transmitting entangled qubits over quantum channels. However, due to the collapse-after-measurement phenomenon, entanglement swapping is realized at the cost of destroying entanglements. Therefore, the entangled pairs shared by adjacent quantum nodes, also called *entanglement links*, are the crucial resource for end-to-end entanglement distribution.

However, two inherent features of quantum networks contribute to the scarcity of entanglement links: (i) *The capability of one quantum node to establish entanglement links with its neighbors is limited*. The success probability of link-level entanglement distribution decreases exponentially with the physical length of a quantum channel. Consequently, it is hard to create entanglement links between adjacent quantum nodes; (ii) *Quantum decoherence introduced by environmental noise results in a reduced number of entanglement links*. Quantum decoherence will break the quantum features of entangled systems, thus reducing the number of available entanglement links. Although entanglement purification [8] has been proposed to alleviate the attenuation of entanglement quality, it works at the cost of consuming entanglement links. Therefore, entanglement links are a scarce resource

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Zhonghui Li, Jian Li, Kaiping Xue, Lutong Chen, Nenghai Yu, and Qibin Sun are with the School of Cyber Science and Technology, University of Science and Technology of China, Hefei 230027, China (e-mail: lijian9@ustc.edu.cn; kpxue@ustc.edu.cn).

Jun Lu is with the School of Cyber Science and Technology and the Department of Electronic Engineering and Information Science, University of Science and Technology of China, Hefei 230027, China.

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in quantum networks. The scarcity of entanglement links inevitably results in a gap between entanglement resources and source-destination (SD) pairs' demands in bottlenecks, thus hindering the implementation of concurrent end-to-end entanglement distributions. To boost request service capability, an end-to-end entanglement distribution design that can effectively deal with the issue of entanglement resource scarcity is required.

In this paper, we focus on the design of end-to-end entanglement distribution in such a general scenario where multiple SD pairs with different demands simultaneously request to establish end-to-end entanglements. The existing work mainly focuses on selecting paths and cannot effectively solve the bottleneck problem caused by the scarcity of entanglement resources. Here, we propose an end-to-end entanglement distribution design called NarrowGap that can overcome bottleneck limitations and improve the request service capability of quantum networks. NarrowGap consists of two essential schemes, i.e., the resource allocation scheme (RAS) and the resource transfer scheme (RTS). RAS guarantees the fairness of entanglement resource allocation by adopting a weighting-based strategy. This strategy considers two differences between different requests: the success probability of link-level entanglement distribution and each SD pair's demand. At the heart of NarrowGap, RTS selects supplementary paths to narrow the gap between the limited entanglement resources and SD pairs' demands in bottlenecks based on the finding that entanglement swapping can not only extend entanglement distribution distances but also realize the "transfer" of entanglement resources. In this way, the supplementary entanglement resources can be used to mitigate resource scarcity in bottlenecks, thus improving the request service capability of quantum networks.

We implement NarrowGap in a system-level quantum network simulation platform named SimQN [9], which can present complete quantum properties and support large-scale network simulation. Extensive simulations show that NarrowGap outperforms the three representative schemes, i.e., resource allocation based on demand proportion, resource allocation using a fair-share manner, and resource allocation based on the success probability of end-to-end entanglement distribution. Hereafter, three schemes are referred to as demand-based, fair-share, and probability-based. NarrowGap shows high fairness, almost similar to the fair-share scheme. Moreover, NarrowGap performs better than three representative schemes in terms of request service rate and can completely meet SD pairs' demands in particular situations. In summary, this paper makes the following contributions:

- We design NarrowGap to deal with the bottleneck problem and thus improve quantum networks' performance in request service capability. RAS proposed in NarrowGap guarantees the fairness of resource allocation, using a weight determined by the success probability of end-to-end entanglement distribution and SD pairs' demands.
- RTS is introduced in NarrowGap to narrow the gap between the limited entanglement resources and requests' demands in bottlenecks, using the unique feature of entanglement swapping. RTS formulates the resource

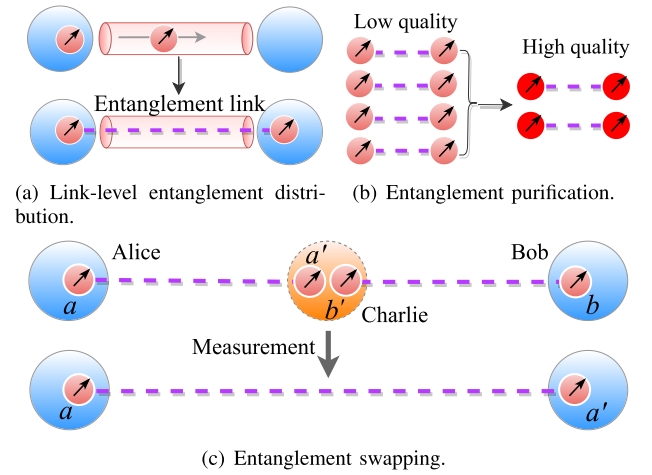


Fig. 1. Three essential quantum operations required for end-to-end entanglement distribution in quantum networks.

allocation on supplementary paths as a multi-dimensional knapsack problem and then presents an approximation algorithm to solve it to maximize the request service rate.

- We evaluate NarrowGap using a discrete-event-based quantum network simulator and compare it with three representative resource allocation schemes in both a randomly generated topology and a real-world network topology, demonstrating the advantages of NarrowGap.

The rest of the paper is organized as follows. Section II mainly introduces the background and motivation of our work. Section III presents the overview of NarrowGap and elaborates on two essential schemes, RAS and RTS. Section IV presents evaluation results, and our work is concluded in Section V.

II. BACKGROUND

This section first reviews three essential quantum operations required for end-to-end entanglement distribution. Then, related work and a motivation example are introduced. Finally, we present the system model and some assumptions.

A. End-to-End Entanglement Distribution

Entangling distant quantum end nodes is pivotal to quantum networks. Fig. 1 depicts three essential quantum operations required for end-to-end entanglement distribution.

1) *Link-Level Entanglement Distribution*: This quantum operation aims to distribute Bell-state entangled pairs between adjacent nodes to establish entanglement links. Generally, the single-atom excitation scheme can be used to achieve this goal pairs [10]. As shown in Fig. 1(a), two adjacent nodes fitted with an optical cavity can share entangled pairs by transmitting entangled photons from one node to another. Concretely, Alice transfers the internal state of an atom to the optical state of a cavity mode employing a laser beam. Then, the atom-entangled photon travels along a quantum channel and enters another cavity at Bob. In Bob's cavity, the photon is absorbed coherently, and its polarization is mapped onto the internal state of an atom. As a result, two atoms located at Alice and Bob are entangled, i.e., an entanglement link is established. Notably, it is challenging to distribute entangled pairs between

two nodes directly linked over long distances due to the inherent loss in quantum channels. The success probability of distributing an entangled pair between adjacent nodes u and v decreases exponentially with the physical length of the attenuated quantum channel, i.e., $p_g \approx e^{-\alpha l_{(u,v)}}$ [11], where α is a constant determined by the physical medium and $l_{(u,v)}$ is the length of the attenuated quantum channel.

2) *Entanglement Purification*: In quantum mechanics, the loss of quantum features or information during the interaction of a quantum system with a noisy environment is known as quantum decoherence, which is one of the key obstacles in the applications of quantum information. Generally, the degree of quantum decoherence can be reflected by *fidelity* [12] used to measure the information overlap between two quantum states, especially the decoherent and initial states. Suppose the initial pure state is $|\psi_1\rangle$ and the another state is ρ_2 . The fidelity F can be formulated by the expression $F \triangleq \langle \psi_1 | \rho_2 | \psi_1 \rangle$, and it satisfies the inequality $0 \leq F \leq 1$. Notably, high-fidelity entanglement facilitates the performance improvement of quantum applications, especially quantum teleportation. Hence, it is required to “recover” as much of the lost quantum information as possible during end-to-end entanglement distribution. In this context, entanglement purification is proposed to mitigate the negative impacts of quantum decoherence and improve entanglement fidelity.

Entanglement purification is an essential quantum technique used to counteract the negative impacts of noise and improve the fidelity of entangled systems. As shown in Fig. 1(b), entanglement purification works by distilling high-quality entangled pairs from low-quality entanglement ensembles. Entanglement purification can be regarded as a quantum operation with input and output, e.g., the CNOT-gate-based purification scheme [13], [14], [15] and the linear-optical-elements-based purification scheme [16], [17]. Generally, entanglement purification uses two low-fidelity entangled pairs as input and finally outputs an entangled pair with high fidelity. That is, entanglement purification improves fidelity at the expense of a reduced number of low-fidelity entangled pairs. Consequently, entanglement purification further contributes to the scarcity of entanglement resources in quantum networks.

3) *Entanglement Swapping*: Entanglement swapping is essentially a local measurement involving the transfer of entanglement to two qubits that were produced independently and never previously interacted. Note that the implementation of entanglement swapping usually follows a probability that depends on the states of two swapped entangled pairs [18]. To explain the implementation of entanglement swapping more intuitively, we here assume this success probability is one. As shown in Fig. 1(c), Alice and Bob deliver an entangled qubit, a' and b' , separated from the local entangled pair to Charlie, respectively. After Charlie performs a local measurement on a' and b' , the quantum information of a' will be projected on b . Consequently, the two remaining qubits located at Alice and Bob are entangled, i.e., Alice establishes a two-hop entanglement with Bob. If the fidelity loss introduced by the local measurement can be negligible, the fidelity of the two-hop entanglement is proportional to the product of the fidelity of two measured entangled pairs.

Notably, entanglement links are broken after entanglement swapping due to the collapse-after-measurement phenomenon. Hence, entanglement swapping can also be regarded as a process in which entanglement resources are transferred from two pairs of adjacent quantum nodes to Alice and Bob.

The three quantum operations mentioned above are essential for end-to-end entanglement distribution. Specifically, link-level entanglement distribution provides the fundamental resources for establishing end-to-end entanglements, entanglement purification plays a pivotal role in improving the quality of end-to-end entanglement, and the extension from short-distance link-level entanglement to end-to-end entanglement is realized with the help of entanglement swapping.

B. Related Work

As a promising platform for quantum applications, one of the essential problems in quantum networks is establishing entanglement between distant quantum end nodes. For this problem, Li et al. [19] proposed two strategies, i.e., connection-oriented and connectionless, inspired by classical communications. For the connection-oriented strategy, an exclusive quantum connection is first built for each SD pair by assigning dedicated memory units along the selected path; then, entangled qubits are distributed on demand to establish dedicated entanglement links; finally, swap operations are performed to generate end-to-end entanglements. The connectionless strategy does not need to establish exclusive quantum connections for SD pairs. Specifically, entanglement links are pre-established as the underlying network resource; then, a path is selected for an SD pair; finally, entanglement resource allocation and swap operations are performed to realize end-to-end entanglement distribution. Compared with the connection-oriented strategy, the connectionless strategy can realize a high end-to-end entanglement distribution rate but with low reliability due to the lack of dedicated entanglement links.

For the connection-oriented strategy, Van Meter et al. [20] studied applying Dijkstra algorithm to quantum networks; Pirandola [21] proposed multi-path routing in a diamond topology; Pant et al. [22] proposed a greedy solution for path selection in grid networks; Shi and Qian [23] proposed a design called Q-CAST for concurrent end-to-end entanglement distributions; Zhao and Qiao [24] proposed a routing design called REPS to maximize the network throughput using the network flow theory; Santos et al. [25] presented an algorithm to find the shortest path that can assure the fidelity of end-to-end entanglement above a specified threshold; Bugalho et al. [26] proposed a multi-objective shortest-path-based algorithm for generating multipartite entanglement between different nodes with noisy quantum repeaters and imperfect memories.

For the connectionless strategy, most existing studies focus on a specific network topology such as a ring or grid [27], [28]; In [29], the authors presented three representative entanglement resource allocation schemes for concurrent end-to-end entanglement distribution requests; Chakraborty et al. [30] proposed two distributed routing algorithms, i.e., the modified

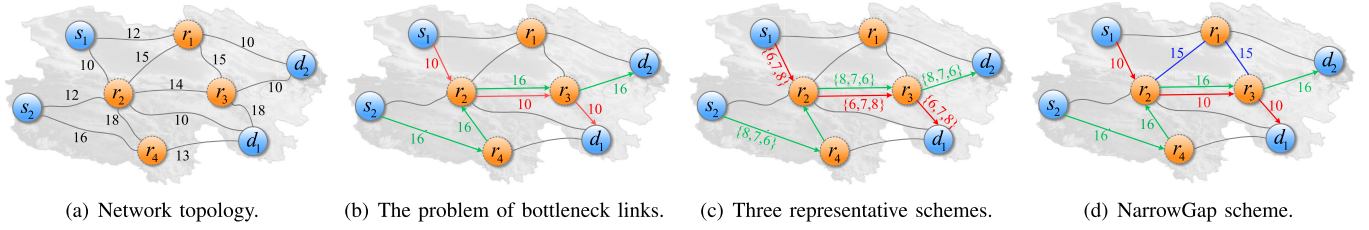


Fig. 2. An example of our work. In subfigure (a), the black edges are quantum channels, and the black numbers present the number of entanglement links over each edge after entanglement purification. In subfigure (b), the green and red edges are SD pairs' requests, and the colored numbers represent each SD pair's demand. In subfigure (c), the set labels represent the number of entanglement links allocated to SD pairs using three resource allocation schemes. In subfigure (d), the blue edges are the supplementary edges, and the blue numbers are the number of supplementary entanglement links.

greedy and local best effort algorithms, to minimize the latency of end-to-end entanglement distribution in quantum networks; Li et al. [31] proposed an entanglement routing algorithm, which guarantees that the fidelity of the established end-to-end entanglement connections can meet the requirements of the SD pairs. Chen et al. [32] designed a decentralized hop-by-hop entanglement distribution scheme and a decentralized routing algorithm that can probe and avoid network congestion in quantum networks.

The existing work mentioned above mainly focuses on path selection and maximizing the utilization of entanglement resources on the selected path. However, they ignore the realistic demands of different SD pairs and fail to overcome the limitations of bottlenecks on quantum networks' request service capability. In this paper, we propose an end-to-end entanglement distribution design to improve the request service capability based on the finding that additional swap operations can be performed on supplementary paths to narrow the gap between the available entanglement resources and the requests' demands in bottlenecks.

C. A Motivation Example

As shown in Fig. 2, we present the motivation of NarrowGap with an example. Fig. 2(a) depicts a general network topology consisting of four quantum end nodes (i.e., s_1 , s_2 , d_1 , and d_2) and four quantum routers (i.e., r_1 , r_2 , r_3 , and r_4). Adjacent quantum nodes are linked via quantum channels used to transmit entangled qubits. Unlike the quantum repeater, which has only a single pair of input and output ports, the quantum router is an extension of the quantum repeater with multiple pairs of input and output ports. That is, quantum routers cannot only extend the distance of entanglement distribution but also route the input entanglement distribution flow to the appropriate output port based on routing results. In order to explain our motivation more clearly and intuitively, the success probability of entanglement swapping is assumed to be 0.8, 0.9, 0.9, and 0.8 at four quantum routers, respectively. Besides, each edge holds more than one usable entanglement link with fidelity above a specific value after entanglement purification, e.g., (r_1, r_2) holds 14 usable entanglement links.

Fig. 2(b) shows that two SD pairs, $\langle s_1, d_1 \rangle$ and $\langle s_2, d_2 \rangle$, request to share 8 and 10 end-to-end entanglements, respectively. Besides, two paths $s_1 \rightarrow r_2 \rightarrow r_3 \rightarrow d_1$ and $s_2 \rightarrow r_4 \rightarrow r_2 \rightarrow r_3 \rightarrow d_2$ are selected for them, respectively.

Notably, in order to satisfy each SD pair's demand, the number of entanglement links allocated by each edge in the selected path theoretically equals the upper bound of the quotient of the SD pair's demand divided by the success probability of end-to-end entanglement distribution due to the probability feature of entanglement swapping. For example, each edge of the selected path needs to allocate 10 (i.e., $\lceil 8 / (0.9 * 0.9) \rceil = 10$) entanglement links to $\langle s_1, d_1 \rangle$. However, the number of available entanglement links held by (r_2, r_3) is less than the total resource demands of two SD pairs. As a result, the edge (r_2, r_3) becomes a bottleneck edge, and there is an inevitable resource competition between two SD pairs.

Three representative resource allocation schemes can handle the resource competition problem in (r_2, r_3) : (i) demand-based scheme, i.e., resource allocation is realized based on the proportion of each SD pair's demand. Consequently, two SD pairs obtain roughly 8 and 6 entanglement links in (r_2, r_3) , respectively; (ii) fair-share scheme, i.e., entanglement links are allocated to SD pairs using a fair share manner, and each SD pair can get seven entanglement links; (iii) probability-based scheme, i.e., each SD pair's ability to successfully establish an end-to-end entanglement along the selected path determines the number of entanglement links allocated to it by (r_2, r_3) . When each edge possesses one entanglement link, the success probability of end-to-end entanglement distribution for the first SD pair equals the product of the success probability of the swap operation performed by each quantum router, i.e., 0.648, while the second is 0.81. Hence, two SD pairs can get 6 and 8 entanglement links in (r_2, r_3) , respectively. Fig. 2(c) shows the set of the number of entanglement links allocated by (r_2, r_3) to two SD pairs is $\{6, 7, 8\}$ and $\{8, 7, 6\}$, respectively, using different resource allocations schemes. Although three schemes can effectively handle the resource competition in (r_2, r_3) , none of them can completely meet two SD pairs' demands due to the limitation of resource scarcity.

In this work, we propose NarrowGap to narrow the gap between the available entanglement resources and SD pairs' demands to mitigate resource competition. NarrowGap is designed based on the finding that the number of entanglement links established by two adjacent nodes can be transformed from less to more by performing additional swap operations, the essence of which is "transfer" idle entanglement links on supplementary paths to bottleneck edges. As shown in Fig. 2(d), 15 idle entanglement links on the path $r_2 \rightarrow r_1 \rightarrow r_3$ can be transferred to (r_2, r_3) by performing swap operations

at r_1 . Notably, 15 entanglement links cannot be completely transformed into supplementary resources due to the imperfection of swap operations. The number of supplementary entanglement links theoretically equals $\lfloor 15 * 0.8 \rfloor = 12$. As a result, (r_2, r_3) possesses 26 entanglement links, equaling the total resource demands of two SD pairs. Hence, two requests can be completely satisfied at the same time, and $\lfloor 16 * 0.648 \rfloor + \lfloor 10 * 0.81 \rfloor = 18$, end-to-end entanglements can be established. NarrowGap can facilitate the implementation of concurrent end-to-end entanglement distribution and performs well in request service capability.

D. System Model

1) *Network Model*: A quantum network can be modeled as an undirected graph. The set of vertices corresponds to quantum nodes, including end nodes, repeaters, and routers. Each node presents the ability to generate, transmit, store, and process qubits, as well as classical bits. Each repeater or router realizes entanglement swapping with a constant probability, and different repeaters or routers vary in this probability. Besides, all quantum nodes can communicate with each other via the existing Internet. The set of edges corresponds to quantum channels, e.g., optical fibers and free space channels. With the help of quantum channels, any pair of adjacent nodes can establish entanglement links with a probability that decreases exponentially with the physical length of the quantum channel. To improve quantum networks' robustness, entanglement links are established in advance in this work, and adjacent quantum nodes can hold multiple entanglement links. Each entanglement link can be identified by two adjacent nodes and the serial numbers of the two memory units used to store an entangled pair.

2) *Time Slot*: For multi-hop parallel entanglement swapping, all repeaters or routers on a path must simultaneously be entangled with their predecessor and successor. However, the physical mechanisms underlying entanglement distribution are stochastic. Although quantum memory has been developed to weaken the effects of the probability feature on multi-hop entanglement swapping, the coherent time and size of quantum memory are severely constrained. Consequently, the time synchronization mechanism is necessary for harmonizing the randomly generated entangled pairs. This paper adopts a time slot design to loosely synchronize time. Each time slot is set to a constant duration determined by quantum devices, and the duration can guarantee that quantum networks establish at least one end-to-end entanglement for each SD pair. Besides, the established end-to-end entanglements in a time slot are used to transmit qubits to support quantum applications, and the remaining end-to-end entanglements required by SD pairs can be established in consecutive time slots.

3) *Centralized Controller*: Generally, there are complex and frequent classical interactions between quantum nodes during end-to-end entanglement distribution since quantum operations require the assistance of classical communications. In this context, local information processing will contribute to the non-negligible latency. To reduce the adverse effects of quantum decoherence, the latency of classical communications must be reduced. The huge benefits of software-defined

networking [33] provide a feasible solution. Hence, this work adopts a centralized control design. The centralized controller is responsible for collecting network information and controlling auxiliary protocols, and more controllers can be adopted with the expansion of network size. In a time slot, the centralized controller first collects network information (e.g., network topology and entanglement swapping probability) and then runs auxiliary protocols to implement end-to-end entanglement distribution. This process will not introduce a high communication latency. First, some network information will be constant in several consecutive time slots. In this, the time spent collecting network information can be significantly reduced. Second, quantum operations can be performed in parallel, which can effectively reduce redundant classical communications. Hence, the centralized controller facilitates the implementation of end-to-end entanglement distribution. In each time slot, the failed quantum operations need to be fed back to the centralized controller, and the number of end-to-end entanglements established in a time slot can be calculated based on feedback results.

E. Assumptions

In this paper, there are some assumptions listed as follows.

1) *Quantum Entanglement*: A significant problem in the process of entanglement distribution is that the entangled pair finally shared by two adjacent quantum nodes is not the perfect Bell state due to the noise errors introduced by quantum channels and quantum operations. First, the qubits are known to experience dephasing even if they are long-lived. Besides, the preparation, transmission, and detection of the entangled pairs may not be perfect. Such errors cannot be overcome by repeated attempts at distributing entangled pairs. Here, we characterize any noise as complete depolarization, which is the worst-case assumption of noise compared to the bit flip error model. After suffering from noise, the entangled state can be written as the Werner state ρ [34], [35]

$$\rho = W|\psi^-\rangle\langle\psi^-| + \frac{1-W}{4}\mathbf{I}_4, \quad W \in [0, 1]. \quad (1)$$

$|\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$ and $\mathbf{I}_4$ denotes the 4×4 identity matrix. W is the Werner parameter indicating the degree of entanglement, and it can completely determine a Werner state. The fidelity of ρ can be written as $F \triangleq \frac{3W+1}{4}$. Hereafter, we assume all entangled pairs are Werner states before and after quantum operations such as entanglement swapping.

2) *Noise Model and Fidelity*: The initial fidelity of an entanglement link is mainly affected by a depolarization process with probability during the transmission of entangled qubits [36]. Hence, the initial fidelity is determined by the probability of depolarization on entering a quantum channel, the probability of depolarization per kilometer of the quantum channel, and the channel length. In this work, we mainly consider the latter two factors to determine entanglement fidelity. More concretely, the fidelity decay model is an exponential decay model, i.e., $F \approx F_0 \cdot e^{-\alpha l}$, where F_0 is the initial fidelity, α is the decay parameter determined by the transmission medium and l is the channel length. Besides, additional fidelity loss will also be introduced during entanglement swapping.

Such fidelity loss is mainly correlated with the reliability of Bell state measurement and two-qubit gate operation. We also assume that quantum repeaters or routers behave identically in terms of these operations' reliability. Generally, most quantum memories lose coherence with time due to dephasing. Hence, the noise of quantum memory is also considered. The Werner states that are stored in quantum memories decohere over time with the following noise model: a Werner state ρ residing in quantum memory for a time Δt will transform into ρ' with the Werner parameter $W_{decayed} = W \cdot e^{-\Delta t/T_{coh}}$, where T_{coh} is the joint coherence time of the two memories holding the entangled pair.

3) *Link-Level Entanglement Distribution*: Due to the inherent photon loss in quantum channels and the imperfections of quantum devices, entanglement distribution at the elementary link level is probabilistic. The greater the link generation probability, more idle entanglement links can be used to narrow the resource gap on bottleneck edges to improve the request service rate of quantum networks. In this work, we assume that all quantum nodes behave identically in the physical medium of entanglement preparation devices. As a result, the physical length of the attenuated quantum channel mainly determines the probability of generating entanglement links between adjacent quantum nodes. The established entanglement links can only be used to serve end-to-end entanglement distribution requests within the current time slot and will become invalid in next time slot. Besides, the initial fidelity of entanglement links created over the same edge is identical since entangled pairs are prepared and distributed through the same quantum medium. However, entangled pairs shared by different adjacent node pairs behave differently in fidelity.

4) *Quantum Operations*: Entanglement swapping can be realized using Bell state measurement. This measurement operation will project the quantum state of the intermediate node into one of four Bell states, thus generating a new mixed state. The entanglement quality of the generated state and the success probability of entanglement swapping generally depends on the initial state of the swapped entangled pairs. Besides, there are three methods to control swap operations on a selected path, i.e., sequential (also known as hop-by-hop) [37], [38], nested (also known as doubling) [39], [40], simultaneous [41], [42], and all three methods are compatible with NarrowGap. Notably, the hop-by-hop method will contribute to high-delay end-to-end entanglement distribution due to the probability feature of quantum operations, and the nested method poses challenges in determining the order in which quantum repeater performs swap operations. In order to reduce the operation complexity and make NarrowGap more suitable for the layered design of the quantum Internet architecture, NarrowGap adopts the simultaneous method, i.e., all intermediate nodes simultaneously perform swap operations and send the results to the destination node through classical channels. Suppose a selected path consisting of $n + 1$ quantum nodes, and the fidelity of the end-to-end entanglement generated by performing swap operations is

$$F_{e2e} = \frac{1}{4} + \frac{3}{4} \left(\frac{\xi(4\eta^2 - 1)}{3} \right)^{n-1} \times \prod_{i=1}^n \left(\frac{4F_i - 1}{3} \right), \quad (2)$$

where ξ is the probability of perfect two-qubit gate operation, η is the accuracy of Bell state measurement, and F_i represents the fidelity of the i -th entanglement link.

High-fidelity entanglement is beneficial to quantum applications. In this work, NarrowGap adopts link-level entanglement purification to realize high-fidelity end-to-end entanglement distribution even in a large-scale quantum network. Concretely, link-level entanglement purification is performed after link-level entanglement distribution to “recover” the quantum information lost in noisy channels, thus establishing high-fidelity entanglement links. So far, entanglement purification technology has gradually matured and can effectively address quantum decoherence introduced by practical quantum setups with noise [43]. Here, we adopt the CNOT-gate-based entanglement purification scheme and symmetric purification scheduling strategy [44] to realize entanglement purification. Thus, two entangled pairs, both with F fidelity, are utilized as the input of entanglement purification, and the fidelity of the output entanglement can be written as

$$F' = \frac{F^2 + \frac{1}{9}(1 - F)^2}{F^2 + \frac{2}{3}F(1 - F) + \frac{5}{9}(1 - F)^2} \quad (3)$$

with the probability of the successful purify operation being $F^2 + \frac{2}{3}F(1 - F) + \frac{5}{9}(1 - F)^2$. This formula shows that the number of purification rounds is determined by the fidelity of the purified entangled pair and the target fidelity. In each purification round, two entangled pairs are consumed to improve fidelity. If the output fidelity is larger than a target value, entanglement purification is completed, and the final generated entangled pair is called an usable entanglement link. Otherwise, other purification rounds are required. Besides, the number of consumed entanglement links increases exponentially with the purification rounds using the symmetric purification scheduling strategy.

III. NARROWGAP DESIGN

This section first presents a high-level overview and design goals of NarrowGap. Then RAS and RTS algorithms are introduced in NarrowGap. We first show the main idea of RAS and then present the weight-based resource allocation strategy. Furthermore, we formulate the problem that needs to be solved in RTS and present an approximate algorithm to solve it.

A. NarrowGap Overview

End-to-end entanglement distribution mainly involves five operations, i.e., link-level entanglement distribution, entanglement purification, path selection, resource allocation, and entanglement swapping. Notably, there are several schemes to order these operations to establish end-to-end entanglements, e.g., the order of entanglement purification and path selection can be swapped. In this paper, entanglement links are established and purified in advance as network resources, and we are mainly concerned with the issue of resource allocation. As shown in Fig. 3, the implementation of NarrowGap in a time slot is described using the following five stages:

- (i) **Link-level entanglement distribution stage**: This stage aims to establish entanglement links between adjacent

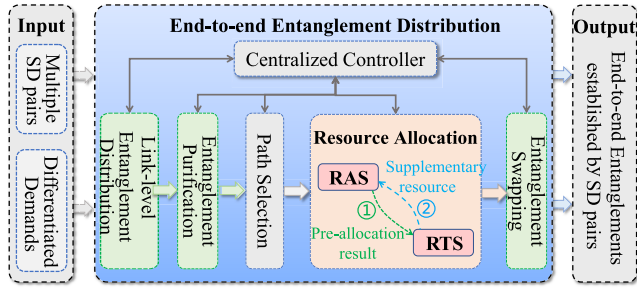


Fig. 3. High-level overview of the NarrowGap design. The heart of NarrowGap is the resource allocation stage consisting of RAS and RTS.

quantum nodes. Each pair of adjacent nodes holds multiple low-fidelity entanglement links after this stage.

- (ii) **Entanglement purification stage:** The second stage aims to improve entanglement fidelity to a certain threshold. After entanglement purification is performed on an edge, the number of entanglement links with fidelity greater than a target value is called the edge capacity. Notably, each edge's capacity is greater than one.
- (iii) **Path selection stage:** At this stage, the centralized controller first collects end-to-end entanglement distribution requests and then determines an identified path for each SD pair using a specific routing algorithm.
- (iv) **Resource allocation stage:** This stage consists of RAS and RTS. RAS determines the number of entanglement links that each edge needs to allocate for requests to solve the resource competition problem. RTS selects supplementary paths to increase bottlenecks' capacity.
- (v) **Entanglement swapping stage:** At this stage, quantum repeaters or routers on selected paths attempt to perform Bell state measurements simultaneously to establish entanglement between distant quantum end nodes.

In a time slot, multiple SD pairs with different demands first initiate end-to-end entanglement distribution requests and can be regarded as the fixed input of NarrowGap, then the above mentioned five stages are performed in sequence and finally outputs end-to-end entanglements. In this paper, we mainly focus on resource competition between multiple requests in bottlenecks. The third stage is the core of NarrowGap. RAS is designed to realize high fairness by using two metrics, i.e., the success probability of end-to-end entanglement distribution and requests' demands, to determine the ratio of resource allocation between multiple requests in bottlenecks. Each edge pre-allocates entanglement links to SD pairs along selected paths according to the results of the RAS algorithm. Then, using the remaining entanglement links possessed by supplementary paths, RTS can increase the number of entanglement links allocated to each request, thus narrowing the gap between the available entanglement link resources and requests' demands. Consequently, a high request service rate and resource utilization can be realized by NarrowGap.

In a time slot, the established end-to-end entanglements are used to support the transmission of qubits and thus support quantum applications. Notably, limited by incomplete quantum setups and memories, scarce entanglement links will hinder the development of quantum networks for a long time.

TABLE I
NOTATIONS LIST

Parameters	Description
(V, E, C)	Network topology. V , E , and C are the set of quantum nodes, quantum channels, and edge capacities in the network topology, respectively.
R	The set of concurrent requests in a time slot.
$C_{(u,v)}$	The capacity of the edge (u, v) after entanglement distribution and entanglement purification.
q_u	The success probability of entanglement swapping at quantum repeater or router $u \in V$.
N_r	The number of end-to-end entanglements that the request $r \in R$ want to establish in a time slot.
p_r	The path selected for the request r in RAS.
$ p_r $	The hop count of the selected path p_r .
Q_r	The success probability of end-to-end entanglement distribution for the requests r .
Q_i^b	The success probability of distributing an entangled pair along the i -th supplementary path selected for the bottleneck edge b in RTS.
$n_{(u,v)}$	The number of requests that share the edge (u, v) .
L_b	the number of entanglement links that edge b lacks.
Variables	Description
$w_{(u,v)}^r$	The entanglement resource allocation weight of the request r on the edge (u, v) .
f^r	The number of entanglement links allocated for the request r along the selected path.
$f_{(u,v)}^r$	The number of entanglement links allocated for the request r on the edge (u, v) .
f_i^b	The number of entanglement links allocated for bottleneck link b along the i -th supplementary path.
J	The Jain's index of resource allocation scheme.
S	The request service rate of quantum networks.
T	The number of end-to-end entangled pairs that can be distributed in a time slot.

Consequently, there will still be some requests that cannot be completely satisfied after RTS, especially in the scenario of high concurrent end-to-end entanglement distribution requests. In order to make our work more complete, NarrowGap can queue the unsatisfied requests into a request pool with a processing priority at the next time slot. In NarrowGap, this priority is mainly determined by two factors: the residual requirement ratio and the waiting time. More concretely, the request with the lowest ratio of the number of end-to-end entanglements that have not been established to its demand has the highest priority; if some requests have the same residual requirement ratio, the request with the longest waiting time will be processed with a higher priority. In this way, the problem of "starvation" can be effectively avoided. With the development of quantum physical devices, we believe the issue that concurrent requests cannot be completely satisfied within a time slot will be effectively solved in the future.

B. Design Goals

For clear presentation, the notations used in this paper are summarized in Table. I. In this paper, we focus on the design of end-to-end entanglement distribution in such a general network scenario: multiple SD pairs with different demands simultaneously request to establish end-to-end entanglements in a time slot. As described in Section I, the scarcity of entanglement links results in resource competition in bottlenecks,

thus hindering concurrent end-to-end entanglement distributions. In this context, the main design goal of NarrowGap is to overcome the limitation of entanglement resource scarcity to improve the request service capability of resource-limited quantum networks.

1) *High Request Service Rate*: The gap between SD pairs' demands and the limited entanglement link resources leads to the fact that a quantum network can only provide partial end-to-end entanglement distribution services for some requests. It is essential for a quantum network to achieve a request service rate that is as high as possible. A high request service rate indicates a high ratio of end-to-end entanglements that can be established to the demands for all requests. Most notably, the path selected for each request with a success probability of end-to-end entanglement distribution. For the request r , this probability is denoted as $Q_r = \prod_{u \in p_r} q_u$, where q_u is the success probability of entanglement swapping at quantum repeater or router u and p_r is the path selected for the request r . The number of end-to-end entangled pairs that can be distributed for the request r in a time slot is the product of f^r and Q_r . Hence, the request service rate S can be written as

$$S = \frac{\sum_{r \in \mathbf{R}} (Q_r f^r / N_r)}{|\mathbf{R}|}, \quad (4)$$

where N_r is the number of end-to-end entanglements that r wants to establish. When all paths are determined, the more entanglement links each edge allocates to each request, the higher the request service rate of quantum networks.

2) *High Fairness*: Due to the scarcity of entanglement links, there are inevitably some bottlenecks due to the gap between the limited entanglement links and concurrent requests' demands. In this context, allocating entanglement links to multiple requests in bottlenecks is considerably challenging. First, each hop on the selected path needs to allocate the same number of entanglement links to each request since swap operations consume entanglement links symmetrically. Therefore, an inappropriate entanglement resource allocation scheme will result in a low request service rate and a high waste of entanglement link resources. In addition, NarrowGap adopts the time slot design to handle multiple requests with differentiated requirements, and the residual requirement ratio mainly determines the processing priority of the input requests. Hence, a poor entanglement resource allocation scheme will harm the stability and reliability of quantum networks. In summary, the solution to the entanglement resource competition problem significantly affects the performance of quantum networks.

In this paper, we aim to design a fairness-guaranteed resource allocation scheme based on the consideration that concurrent end-to-end entanglement distribution requests have differentiated resource demands. High fairness means that the limited entanglement link resources in bottlenecks can be allocated to multiple requests impartially. Here, we adopt Jain's index [45] to evaluate the fairness of the entanglement resource allocation scheme. In this, the fairness J is given by:

$$J = \frac{(\sum_{r \in \mathbf{R}} f^r)^2}{|\mathbf{R}| \sum_{r \in \mathbf{R}} (f^r)^2}, \quad (5)$$

where f^r is the number of entanglement links allocated to the request r along the selected path, and $|\mathbf{R}|$ is the number of end-to-end entanglement distribution requests in a time slot.

C. RAS Design

1) *Main Idea*: RAS is designed to deal with the issue of entanglement resource competition in bottlenecks and achieve high-fairness entanglement resource allocation. The design of RAS is based on the following two core ideas:

Each edge preferentially allocates entanglement links to the request with a high demand for entanglement resources. In this paper, NarrowGap is designed to improve quantum networks' request service capability. On bottleneck edges, end-to-end entanglement distribution requests will differ in the number of end-to-end entangled pairs they want to share. According to the above description, the request service rate is positively related to the sum of the proportion of the available end-to-end entangled pairs to the required end-to-end entanglement per request. Therefore, in order to realize a high request service rate in a time slot, entanglement links are preferentially assigned to the requests with a high demand for entanglement resources. In this way, each SD pair's requirement can be satisfied with a high proportion, thus facilitating the improvement of the request service rate.

Each edge preferentially allocates entanglement links to the request with a high success probability of end-to-end entanglement distribution. After the path selection stage, each request gets a unique path to realize end-to-end entanglement distribution. However, concurrent requests vary in the success probability of establishing an end-to-end entanglement since all the intermediate quantum nodes on the selected path will vary in the success probability of entanglement swapping. To improve quantum networks' request service capability and the number of entangled pairs that can be shared by distant quantum end nodes finally, the RAS scheme preferentially allocates entanglement links to the request with a high success probability of end-to-end entanglement distribution. In this way, NarrowGap can have a good performance in terms of the request service rate and the number of end-to-end entangled pairs that can be distributed in a time slot.

2) *Weight-Based Strategy*: Inspired by the above two ideas, RAS introduces a weight-based strategy to realize entanglement resource allocation in bottleneck edges. The weight directly indicates the proportion of the entanglement link resources allocated to an end-to-end entanglement distribution request to the total available entanglement link resources. The weight-based strategy combines the insights of probability-based and demand-based schemes, i.e., the weight is determined by two important factors: requests' demands for entanglement links and the success probability of end-to-end entanglement distribution. For the request r , the success probability of end-to-end entanglement distribution along the selected path p_r is Q_r , and the number of entanglement links required by r is denoted as N_r . We assume that the edge (u, v) on the selected path is shared by $n_{(u,v)}$ requests. To balance the influence of the two metrics on the fairness of entanglement resource allocation, we use the product of the percentage of the two factors for each request as the weight

Algorithm 1 RAS Algorithm

Input: The set of requests $\mathbf{R}$, the selected paths;
Output: The number of entanglement links allocated to each request along the selected path;

- 1 Initialize the number of entanglement links allocated to the request r along the selected path p^r , i.e., $f^r \leftarrow 0$;
- 2 Initialize the set of bottleneck edges, $\mathbf{B} \leftarrow \emptyset$;
- 3 **Step 1: Resource Allocation Weight Calculation**
- 4 **for** each request $r \in \mathbf{R}$ **do**
- 5 **for** the edge $(u, v) \in p_r$ **do**
- 6 **if** $C_{(u,v)} < \sum_{i=1}^{n_{(u,v)}} N_{r_i}$ **then**
- 7 Add the edge (u, v) into $\mathbf{B}$;
- 8 Calculate the weight $w_{(u,v)}^r$;
- 9 **end**
- 10 **end**
- 11 **end**
- 12 **Step 2: Resource Allocation Precalculation**
- 13 **for** any edge $(u, v) \in \mathbf{B}$ **do**
- 14 Normalize all requests' weights in the (u, v) ;
- 15 Calculate $f_{(u,v)}^r$, the number of entanglement links allocated to r in the edge (u, v) ;
- 16 **end**
- 17 **Step 3: Resource Allocation Decision**
- 18 Select the minimum $f_{(u,v)}^r$ as f^r ;
- 19 **return** the entanglement link resource f^r allocated to each request along the selected path.

for entanglement resource allocation. Hence, the weight of the request r on the edge (u, v) can be written as

$$w_{(u,v)}^r = \frac{Q_r}{\sum_{i=1}^{n_{(u,v)}} Q_{r_i}} \cdot \frac{N_r}{\sum_{i=1}^{n_{(u,v)}} N_{r_i}}. \quad (6)$$

Notably, the capacity of the edge (u, v) determines how entanglement links are allocated to multiple requests. If the capacity of the edge (u, v) is equal to or larger than the entanglement resource demands of all requests, i.e., $C_{(u,v)} \geq \sum_{i=1}^{n_{(u,v)}} N_{r_i}$, the number of entanglement links allocated to r is directly determined by N_r . In this, each request's demand in this edge can be fully satisfied. However, if $C_{(u,v)} < \sum_{i=1}^{n_{(u,v)}} N_{r_i}$, i.e., the edge (u, v) is a bottleneck edge, the number of entanglement links allocated to r in the edge (u, v) must be determined by $w_{(u,v)}^r$.

3) *RAS Algorithm*: The implementation of the RAS algorithm mainly consists of three steps and is shown in **Algorithm 1**. In the first step, RAS calculates each request's resource allocation weight in each bottleneck edge before determining the number of entanglement links allocated to r (Lines 3-11). In the second step, based on the resource allocation weight of each request, we can get the proportion of the entanglement resources allocated to the request r to the total entanglement resources by normalizing the weights in each bottleneck edge and thus obtain the number of entanglement links allocated to r in each edge (Lines 12-16). It is worth noting that the available number of entanglement links for each request is determined by the minimum number of entanglement links that can be allocated in the selected path.

This is because the swap operation consumes entanglement link resources symmetrically. Hence, RAS uses the minimum number of entanglement links that can be allocated to r in each edge as the number of entanglement links finally allocated to the end-to-end entanglement distribution request r along the selected path (Lines 17-18). Based on the results of the RAS algorithm, the centralized controller can inform each quantum node in selected paths to pre-allocate entanglement links for requests with the assistance of classical communications.

Theorem 1: In the worst case, the runtime complexity of the RAS algorithm is $\mathcal{O}(m|N|^2 + k(m+1)|N|)$.

Proof: Suppose there are k end-to-end entanglement distribution requests, the average hop count of the selected path is $|N|$, and each hop is shared by m requests on average. At **Step 1**, the runtime complexity can be calculated by $\mathcal{O}(km|N|)$. The worst case in RAS is that each hop is a bottleneck edge, i.e., the number of bottleneck edges is $|N|^2$. At **Step 2**, the worst-case runtime complexity is $\mathcal{O}(m|N|^2)$. At **Step 3**, the worst-case runtime complexity can be calculated by $\mathcal{O}(k|N|)$. Therefore, the worst-case runtime complexity of the RAS algorithm is $\mathcal{O}(m|N|^2 + k(m+1)|N|)$.

D. RTS Design

After RAS, there are idle entanglement links in quantum networks. To improve the request service rate and mitigate network congestion, it is better to make full use of these idle entanglement links. Here, we introduce RTS in NarrowGap. RTS works by selecting supplementary paths to transfer the idle entanglement links to bottleneck edges, thus narrowing the gap between the available entanglement resources and SD pairs' demands. More concretely, the network system takes the two endpoints of each bottleneck edge as an SD pair to select supplementary paths. In order to increase the request service rate as much as possible, we select multiple paths with no overlapping edges or as few overlapping edges as possible for multiple bottleneck edges. Notably, requests' demands may not necessarily be fully satisfied within a time slot by selecting supplementary paths due to the scarcity of entanglement resources, and determining the optimal set of supplementary paths is challenging. Hence, we do not specifically explore how to realize the optimal selection of supplementary paths. In this work, we use a modified shortest path algorithm to select supplementary paths with no overlapping edges for each bottleneck edge. The modified shortest path algorithm mainly uses the success probability of end-to-end entanglement distribution as the routing metric. NarrowGap first selects the shortest path and deletes the selected edges from the network topology, then another path is selected in the new network topology using the shortest path algorithm. These two steps are executed repeatedly until all supplementary paths are determined.

Suppose two supplementary paths with different success probabilities of end-to-end entanglement distribution are selected to supplement entanglement links for the bottleneck edge (r_1, r_4) as shown in Fig. 4. The success probabilities of r_2 and r_3 are 0.8 and 0.6, respectively. Two supplementary paths should expectantly create enough entangled pairs to sum to 3. There are four methods: 5 entanglement links are

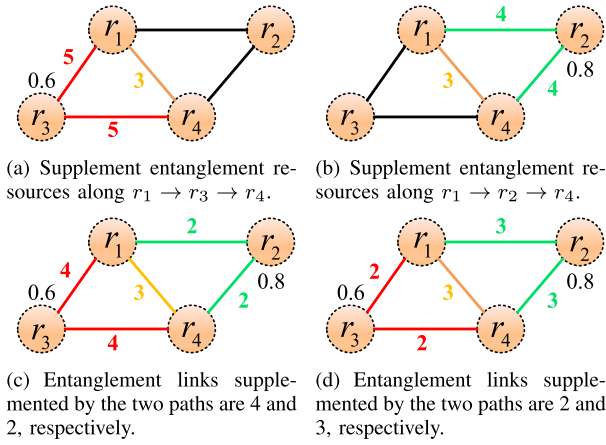


Fig. 4. Four different methods to supplement entanglement links for the bottleneck edge (r_1, r_4) .

allocated on each edge along $r_1 \rightarrow r_3 \rightarrow r_4$ (as shown in Fig. 4(a)); 4 entanglement links are allocated on each edge along $r_1 \rightarrow r_2 \rightarrow r_4$ (as shown in Fig. 4(b)); $r_1 \rightarrow r_3 \rightarrow r_4$ and $r_1 \rightarrow r_2 \rightarrow r_4$ assign 4 and 2 entanglement links, respectively (as shown in Fig. 4(c)); $r_1 \rightarrow r_3 \rightarrow r_4$ and $r_1 \rightarrow r_2 \rightarrow r_4$ assign 2 and 3 entanglement links, respectively (as shown in Fig. 4(d)). Although both methods can improve the capacity of (r_1, r_4) , they vary in the global profits of improving the request service rate.

1) *Problem Formulation:* We summarize the issue of assigning idle entanglement links among multiple supplementary paths to increase bottlenecks' capacity in RTS as a problem to maximize the request service rate. The set of bottleneck edges in a time slot is denoted as $\mathbf{B}$. We assume that the number of supplementary paths selected to realize entanglement resource transfer is k . The resource allocation problem to maximize the request service rate in RTS can be formulated as:

$$\text{Maximize } S = \frac{\sum_{r \in \mathbf{R}} (Q_r f^r / N_r)}{|\mathbf{R}|} \quad (7)$$

subject to:

$$\sum_{i=1}^k f_i^b \cdot Q_i^b \leq L_b, \quad (7a)$$

$$\sum_{b \in \mathbf{B}} f_i^b \leq C_{(u,v)}, \quad (7b)$$

$$f_i^b \text{ is integer, } \forall i, b, \quad (7c)$$

where f_i^b is the number of entanglement links allocated for bottleneck edge b along the i -th supplementary path, Q_i^b is the success probability of multi-hop entanglement distribution of the i -th supplementary path selected for the bottleneck edge b , and L_b is the number of entanglement links that b lacks; and $C_{(u,v)}$ is the capacity of the edge (u,v) in the supplementary paths, i.e., the number of entanglement links established between adjacent quantum nodes u and v . For example, the capacity of the edge (r_1, r_2) in the supplementary path $r_1 \rightarrow r_2 \rightarrow r_4$ is four. For RTS, the entanglement resources allocated by k supplementary paths need to be less than or equal to the number of entanglement links that b

lacks (constraint (7a)), and the number of entanglement links allocated for bottleneck links in any edge must be less than its capacity (constraint (7b)).

Since there is resource competition on the shared edge and different resource transformation strategies contribute to the different profits of improving the request service rate, two problems need to be solved: (i) Which supplementary paths should be selected to serve entanglement resource transformation? (ii) How many entanglement links does each supplementary path need to allocate for each bottleneck edge? Here, the selection of supplementary paths and the assignment of entanglement links are similar to the reverse process of selecting items to place in different knapsacks. Therefore, the entanglement resource allocation among multiple supplementary paths can be regarded as a generalized multidimensional knapsack problem that cannot be solved with a polynomial time [46]. Specifically, the number of entanglement links possessed by each supplementary path can be regarded as the capacity of a knapsack, and the number of entanglement links missing by each bottleneck corresponds to the number of items. In this, the problem of maximizing the request service rate can be approximately equivalent to the problem of maximizing benefits by placing items in multiple knapsacks. To solve the problem expressed in Eq. (7), we design an approximation algorithm inspired by [47], as shown in **Algorithm 2**, to maximize the request service rate.

2) *RTS Algorithm:* The RTS algorithm is an approximate one, and it consists of two essential steps: guessing solutions and packing edges. In the RTS algorithm, y is an item of an allocation solution, $s(y)$ is the number of entanglement links that y occupied, $p(y)$ is the profit in the requested service rate brought by the item y , and OPT is the optimal solution. The RTS algorithm first “guesses”, in polynomial time, some feasible solutions to the problem of maximizing the request service rate described in Eq. (7). These solutions guarantee that the profit of improving the request service rate is greater than or equal to $(1 - \epsilon)OPT$ in a polynomial time (Lines 2-11). For each guessed objective solution $\mathcal{O}$, the RTS algorithm starts to structure edge blocks, thus relaxing the strongly NP-complete problem. After that, the RTS algorithm attempts to pack the most profitable items and large items contained in each guessed solution into blocks (Lines 12-20). For the remaining items, all items are packed via a Linear Programming (LP) approach (Lines 21-30). Finally, the RTS algorithm collects the request service rate gain profit brought by all possible solutions and returns the maximum value as the optimal result of resource allocation.

Theorem 2: *RTS algorithm gives a 2-approximation for multidimensional knapsack problem.*

Proof: Let Q be the set of indices i such that B_i is small. Define Q' to be the set of $t = 1/\epsilon + \lceil 4\epsilon^{-1} \ln 1/\epsilon \rceil$ largest indices in the set Q . Let B_Q and $B_{Q'}$ be the sets of all edges in the blocks specified by the index sets Q and Q' , respectively. Let j be the largest index in Q such that $j \leq i - 4\epsilon^{-1} \ln 1/\epsilon$. For any $k \leq j$, an edge in block B_k has capacity at most $1/(1 + \epsilon)^{i-k}$. Thus the edge capacity in B_k is at most $(1 + \epsilon)^{j-i+1}/(1 + \epsilon)^{j-k+1} \leq \epsilon^2/(1 + \epsilon)^{j-k+1}$. Since B_k is a small block, the total capacity of all edges in

Algorithm 2 RTS Algorithm

Input: n solutions, approximation ratio ϵ ;
Output: Resource allocation to maximize S ;

```

1  $p_{max} \leftarrow$  the largest value among solution profits;
2 Guess a feasible solution  $\mathcal{O}$  such that
   $\max\{p_{max}, (1 - \epsilon)OPT\}$ ;
3 Set bounds on  $OPT$ :  $p_{max} \leq OPT \leq n \cdot p_{max}$ ;
4 Discard all solutions  $y$  where  $p\{y\} < \epsilon \cdot \mathcal{O}/n$ ;
5 Divide all profits by  $\epsilon \cdot \mathcal{O}/n$ , and round down to nearest
  power of  $(1 + \epsilon)$ ;
6 Let  $h \leq 4\epsilon^{-1} \ln n + 1$  be the number of distinct profits;
7 Partition all solutions into  $h$  sets, i.e.,  $\{A_1, \dots, A_h\}$ ;
8 Let  $U_i$  be the intersection of the feasible sets and  $A_i$ ;
9 if  $p(U_i) \leq \epsilon \mathcal{O}/h$  then
10 | Discard the set  $A_i$ ;
11 end
12 for each guessed solution do
13 | Order edges in increasing order of their capacity;
14 | Partition them into blocks  $B_0, B_1, \dots, B_\ell$ 
   $(1 \leq \ell \leq n^{O(1/\epsilon^3)})$ ;
15 |  $m_i \leftarrow$  the number of edges in block  $B_i$ ;
16 |  $C_i \leftarrow$  the capacity of the edge in which  $y$  is packed;
17 | Packing the most profitable items into small edges
  blocks with  $m_i \leq 1/\epsilon$ ;
18 | if  $s(y) \geq \epsilon \cdot C_i$  then
19 | | Packing  $y$  (a large item) into large edges blocks
  with  $m_i > 1/\epsilon$ ;
20 | end
21 | while  $m_i$  not equals  $n$  do
22 | | if  $s(y) \leq \epsilon \cdot (1 + \epsilon)^i$  then
23 | | | Assign items  $y$  to an edge in a large block;
24 | | end
25 | | if  $p(y) \leq \frac{\epsilon}{1+\epsilon} p(Y_j)$  then
26 | | | Assign items  $y$  to an edge in a small block;
27 | | end
28 | |  $m_i \leftarrow m_i + 1$ ;
29 | end
30 end
31 return the solution  $y$ , and the maximize value of  $\mathcal{O}$ 

```

B_k is at most $\epsilon/(1 + \epsilon)^{j-k+1}$, and the total capacity of all edges in small blocks with indices less than or equal to j is $\sum_{k \leq j} \epsilon/(1 + \epsilon)^{j-k+1}$. Therefore, we can pack all items in B_k with $k \in B_Q, k \leq j$. Each of the $1/\epsilon$ largest edges in B_Q could be in their own blocks. Hence, the largest t indices from Q would contains all these blocks. Hence, there exists a set $U' \subseteq U$, where U is the set of solutions that can be packed in the edges B_Q , such that U' can be packed into the bins $B_{Q'}$, and $p(U') \geq (1 - \epsilon) \cdot p(U)$. As a result, we have a feasible guess solution and profit at least $(1 - \epsilon)OPT$ in the guessing and structuring the edges steps. In the packing step, if an item's profit is at the least half the profit of the edge in which the item is packed, we retain it and leave out the rest items in the edge it packed in. Otherwise, this item is removed out. This results in a feasible solution of at least

half the profit given by the LP solution. Hence, RTS gives a 2-approximation. ■

IV. PERFORMANCE EVALUATION

This section starts by describing the simulation setup and then simulation results are presented to demonstrate the advantages of NarrowGap.

A. Simulation Setup

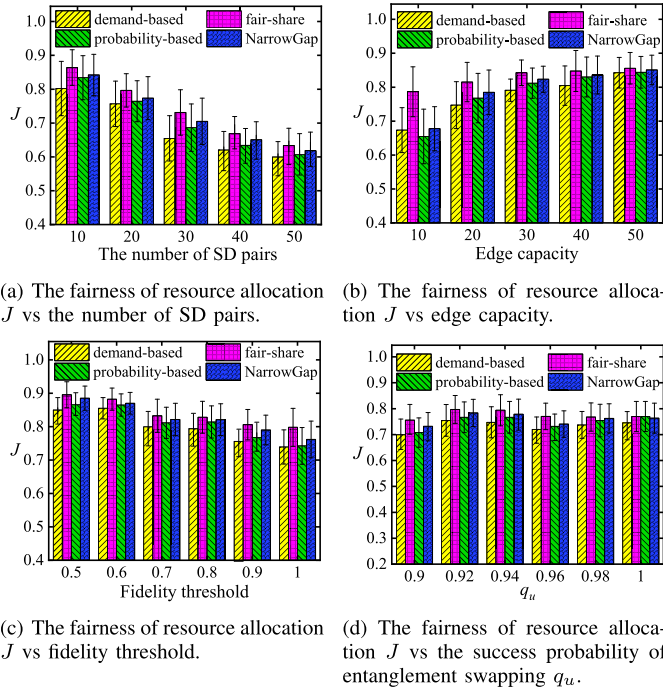
1) *Methodology*: We evaluate the performance of NarrowGap using a discrete-event-based simulation platform named SimQN. Most of the existing work uses the Waxman model or the Erdős–Rényi model to generate a random quantum network [23], [25], [48], [49]. However, the network topology generated based on these models will not give a more realistic presentation of real-world networks. In this paper, we randomly place 100 nodes into a 10^3km by 10^3km square area using the uniform distribution and adopt the Salama model [50] to determine the probability that nodes u and v are directly linked. The Salama model is an improvement model on the Waxman model and can reproduce many real-world network properties. The connectivity probability of nodes u and v can be expressed by

$$p_{(u,v)} = \frac{k\bar{e}}{|V|} \gamma e^{-l_{(u,v)}/\beta \cdot L_{max}}, \quad (8)$$

where $\bar{e}$ is a parameter used to control the average node degree, k is a constant, $|V|$ is the size of nodes, γ controls the ratio of the long edge to the short edge, β controls the degree of the generated network, and L_{max} is the largest distance between any pair of nodes. Besides, the success probability of establishing an entanglement link is determined by the expression $p_g \approx e^{-\alpha l_{(u,v)}}$. In a fixed network topology, the parameter α determines the success probability p_g . After some testing, we set α to 2×10^{-4} to ensure p_g is greater than 0.5, which is conducive to the analysis of simulation results. In addition, SD pairs and their demands are randomly determined once at the beginning of each time slot to explore the performances of NarrowGap. Through numerous simulations, we show the average results and standard deviations.

2) *Comparison Schemes*: Although the three representative schemes, i.e., demand-based, fair-share, and probability-based, can tackle the issue of entanglement resource competition in bottleneck edges, neither considers the unique features of entanglement swapping. NarrowGap introduces RAS to guarantee the fairness of entanglement resource allocation. Besides, RTS is introduced to increase bottleneck edges' capacity, thus improving quantum networks' performances in terms of the request service rate. We compare these three schemes with NarrowGap under various simulation settings, including fidelity threshold, edge capacity, q_u (i.e., the probability that the repeater or router u successfully performs entanglement swapping), and the number of SD pairs.

3) *Performance Metrics*: We compare NarrowGap with three representative schemes using three evaluation metrics. One is the request service rate S expressed as Eq. (4), which directly reflects quantum networks' request service capability.

Fig. 5. The effects on the fairness of resource allocation J .

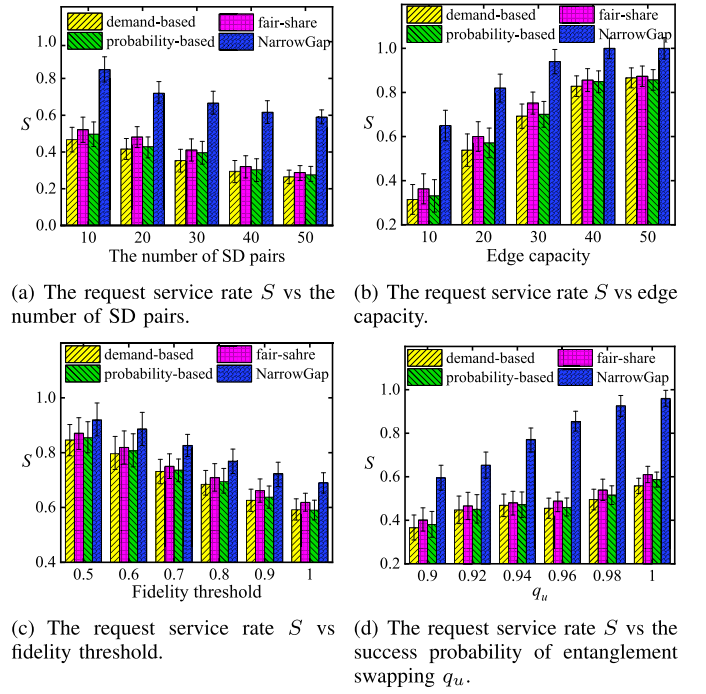
The second metric is the fairness of entanglement resource allocation J determined by Eq. (5). A larger Jain's index means a better fairness performance of the resource allocation scheme. The last is the end-to-end entanglement distribution rate (EDR), i.e., the number of entangled pairs that can be distributed between distant quantum end nodes per time slot, and its unit is end-to-end entangled pairs per time slot. The greater the number of end-to-end entanglements output by NarrowGap, the more entanglement resources RTS will supplement for bottleneck edges after RAS.

B. Simulation Results

1) *Main Observations*: Extensive simulations show that NarrowGap can effectively facilitate the implementation of concurrent end-to-end entanglement distributions in quantum networks. Besides, NarrowGap outperforms three representative schemes in request service rate and the number of end-to-end entangled pairs while showing high fairness. The fairness of NarrowGap is close to the fair-share scheme. In addition, NarrowGap increases the number of end-to-end entangled pairs that can be distributed in a time slot by 68.1%. In particular, the request service rate of NarrowGap can approximately reach 100% in resource-unlimited scenarios.

2) *Fairness*: Fig. 5 shows how four factors, i.e., the number of SD pairs, edge capacity, fidelity threshold, and the success probability of entanglement swapping, affect the fairness performances of different schemes.

a) *Effect of the number of SD pairs*: To evaluate the effects of the number of SD pairs on fairness, we set the success probability of entanglement swapping as 0.9, the average edge capacity is 20, and the fidelity threshold is 0.9. As shown in Fig. 5(a), the fairness performances of the four schemes decrease as the number of SD pairs increases due

Fig. 6. The effects on the request service rate S .

to the increasing competition for entanglement resources. The fair-share scheme performs well in fairness because it treats all requests equally. Although NarrowGap does not outperform the fair-share scheme, the fairness performance of which is quite close to the fair-share scheme.

b) *Effect of edge capacity*: From Fig. 5(b), we can make the opposite observation as we investigate the effects of the number of SD pairs. That is, the fairness performances of the four schemes slowly increase, and the gap between them gradually decreases when the edge capacity increases. This is because the increase of edge capacity can effectively alleviate the competition for entanglement resources in bottlenecks.

c) *Effect of fidelity threshold*: When the fidelity threshold varies from 0.5 to 1.0, the fairness performances of different schemes are shown in Fig. 5(c). These simulation results are similar to the effects of the number of SD pairs on fairness. This figure shows that the fairness performances of the four schemes slightly decrease when the fidelity threshold increases. The reason is that the larger the fidelity threshold, the lower the capacity of each edge.

d) *Effect of the success probability of entanglement swapping*: To investigate how the success probability of entanglement swapping affects the fairness performances of different schemes, we vary this probability from 0.9 to 1.0. These simulation results are shown in Fig. 5(d). From this figure, we can investigate that the success probability of entanglement swapping barely affects fairness. This is because a slight increase in the probability has little effect on the probability of end-to-end entanglement distribution when the network model and the number of SD pairs are determined. Besides, the fairness performance of NarrowGap is consistently close to that of the fair-share scheme.

3) *Request Service Rate*: Fig. 6 shows how different factors affect the request service rate performances of four schemes.

a) *Effect of the number of SD pairs*: Fig. 6(a) shows the effect of the number of SD pairs on the request service rate. We can find that the request service rates of four schemes decrease when the number of SD pairs increases. The larger the number of SD pairs, the greater the resource competition in bottlenecks. As a result, the number of entanglement links allocated to each SD pair gradually decreases. Hence, the request service rate gradually decreases. Besides, NarrowGap always outperforms the other three schemes because it can use idle entanglement resources to increase bottlenecks' capacity. When the number of SD pairs is 50, NarrowGap can achieve up more than twice the performance improvement in request service rate compared to the demand-based scheme.

b) *Effect of edge capacity*: In Fig. 6(b), the request service rate versus the edge capacity is evaluated in such settings: the number of SD pairs is 20, the fidelity threshold is 0.9, and the success probability of entanglement swapping equals 0.9. As we expected, the request service rate of different schemes increases when the edge capacity increases. This is because the improvement in edge capacity can effectively alleviate the resource competition in bottlenecks, thus resulting in an improvement of the request service rate. Besides, NarrowGap outperforms the other three representative schemes in this simulation. However, the superiority of NarrowGap decreases as edge capacity increases. The reason is that the larger the edge capacity, the fewer the number of entanglement links missing from bottleneck edges.

c) *Effect of fidelity threshold*: To investigate how the fidelity threshold affects different schemes' request service rates, we set the number of SD pairs, the edge capacity, and the success probability of entanglement swapping to 20, 0.9, and 0.9, respectively, and the fidelity threshold is varied from 0.5 to 1.0. The corresponding results are shown in Fig. 6(c). The higher the fidelity threshold, the more entanglement links will be consumed for improving fidelity, i.e., the smaller the capacity of bottleneck links. Hence, we can make the same observation as we investigate the effects of the number of SD pairs. That is, the request service rate of the four schemes decreases when the fidelity threshold increases.

d) *Effect of the success probability of entanglement swapping*: When the success probability of entanglement swapping varies from 0.9 to 1.0, the request service rates of four schemes are shown in Fig. 6(d). We can see that the request service rate gradually increases as the success probability of entanglement swapping increases. NarrowGap outperforms the other representative schemes. Besides, the superiority of NarrowGap increases progressively with the success probability of entanglement swapping. This is because the higher the success probability of entanglement swapping, the more idle entanglement resources can be transferred to improve bottlenecks' capacity.

4) *EDR*: Fig. 7 shows how four factors affect the number of end-to-end entanglements finally established per time slot.

a) *Effect of the number of SD pairs*: To investigate the effects of the number of SD pairs on throughput, we vary the number of SD pairs from 10 to 50. These simulation results are shown in Fig. 7(a). With the increase in the number of SD pairs, T will also increase since more entanglement

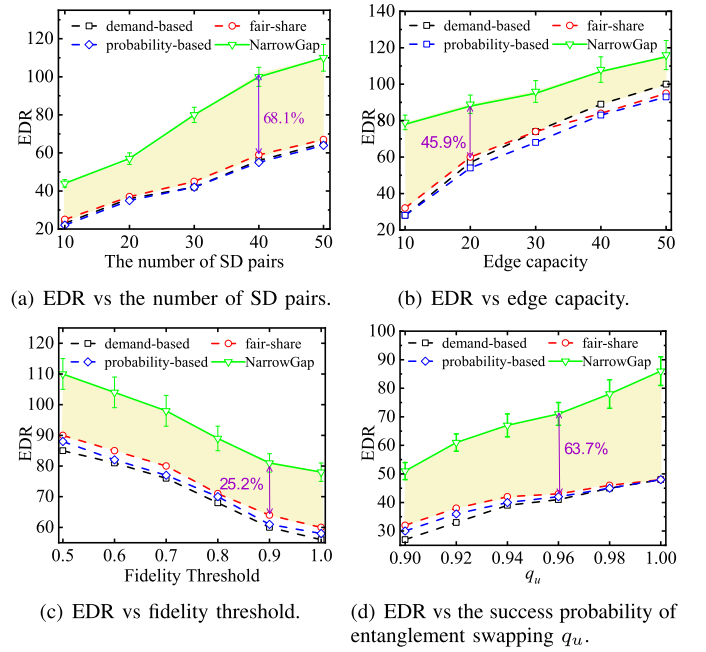


Fig. 7. The effects on the end-to-end entanglement distribution rate EDR.

links are allocated to establish end-to-end entanglements. The three representative schemes perform poorly in EDR due to the limited resources in bottlenecks. However, because NarrowGap can make full use of idle entanglement links, NarrowGap significantly outperforms the other three schemes. When the number of SD pairs equals 40, NarrowGap increases EDR by 68.1% compared to the fair-share scheme.

b) *Effect of edge capacity*: Fig. 7(b) shows how the edge capacity affects EDR. In this simulation, we fix the number of SD pairs, the fidelity threshold, and the success probability of entanglement swapping to 20, 0.9, and 0.9, respectively. The larger the edge capacity, the larger the number of available end-to-end entanglements. This is because the gap between entanglement resources and SD pairs' demands in each bottleneck link is mitigated by improving each edge's capacity. When the average capacity of each edge is equal to 20, NarrowGap can increase EDR by 45.9%. However, the advantage of NarrowGap diminishes as edge capacity increases because the number of the lacking entanglement links in each bottleneck is gradually getting smaller.

c) *Effect of fidelity threshold*: When the fidelity threshold varies from 0.5 to 1.0, the performances of different schemes are shown in Fig. 7(c). From this figure, we can make the opposite observation as we investigate the effects of edge capacity. Specifically, the EDR of the four schemes decreases when the fidelity threshold increases. This is because the larger the fidelity threshold, the smaller the number of entanglement links will be used to improve entanglement fidelity in each edge. In this simulation, NarrowGap still outperforms three representative schemes. When the fidelity threshold is equal to 0.9, NarrowGap can increase EDR by 25.2% compared to the fair-share scheme.

d) *Effect of the success probability of entanglement swapping*: To observe how the success probability of

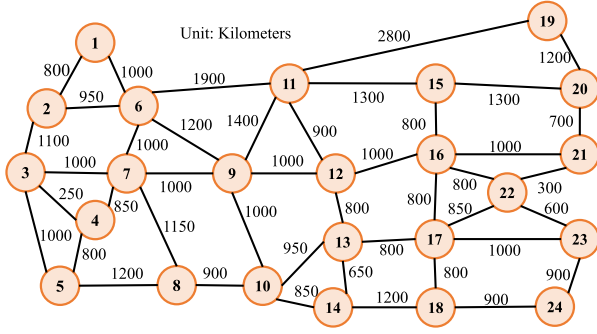


Fig. 8. The real-world network USNET.

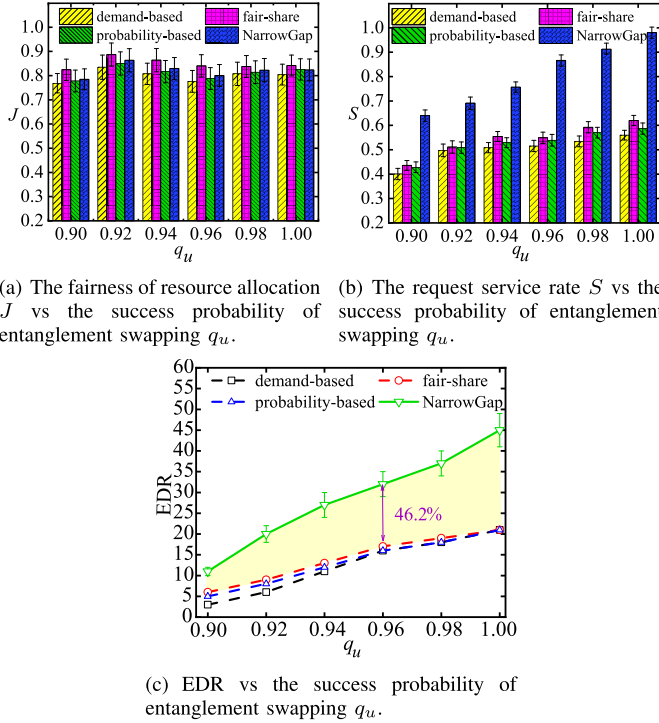


Fig. 9. The performances of NarrowGap in a real-world network.

entanglement swapping affects T , we vary the success probability of entanglement swapping from 0.9 to 1.0. The simulation results are shown in Fig. 7(d). We can see that EDR increases with the improvement of the success probability of entanglement swapping. When the success probability of entanglement swapping approaches 1.0, the performance gap of the three representative schemes gradually approaches zero because the number of entanglement links allocated to each request will gradually equal the number of end-to-end entanglements. Besides, NarrowGap always outperforms the other three representative schemes, and NarrowGap can increase EDR by 63.7% when the success probability of entanglement swapping is 0.96.

C. Implementing NarrowGap in USNET

To fully explore the superiority of NarrowGap, we implement NarrowGap in a real-world network called USNET to compare it with other three schemes. As shown in Fig. 8, the network topology contains 24 nodes and 43 edges, and the

distance between two adjacent nodes is marked on each edge (e.g., the distance between nodes 1 and 2 is 800 km). In this simulation, we assume that there are 5 SD pairs requested to establish end-to-end entanglements and evaluate the effects of the success probability of entanglement swapping on different schemes in terms of fairness, the request service rate, and the end-to-end entanglement distribution rate. As shown in Fig. 9, simulation results are roughly consistent with the above results in randomly generated network topology.

Fig. 9(a) shows that the success probability of entanglement swapping barely affects fairness. Besides, the fairness performance of NarrowGap is consistently close to that of the fair-share scheme. As shown in Fig. 9(b), the request service rate gradually increases as the success probability of entanglement swapping increases. NarrowGap outperforms the other three schemes, and its superiority increases progressively with the success probability of entanglement swapping. As shown in Fig. 9(c), EDR increases with the success probability of entanglement swapping. Besides, NarrowGap always outperforms the other three representative schemes, and NarrowGap can increase EDR by 46.2% when the success probability of entanglement swapping is 0.96.

V. CONCLUSION

In this paper, we proposed NarrowGap to overcome bottlenecks' limitations on concurrent end-to-end entanglement distributions. NarrowGap consists of two essential schemes, RAS and RTS. RAS guarantees the fairness of entanglement resource allocation by considering both the success probability of end-to-end entanglement distribution along the selected path and SD pairs' demands. RTS transfers entanglement resources to narrowing the gap between the limited entanglement resources and SD pairs' demands, thus improving the request service capability of quantum networks. We formulated the problem of resource transfer as a multidimensional knapsack problem and designed an approximation algorithm to solve it. Finally, we compared NarrowGap with three representative schemes in a randomly generated topology and a realistic topology. Extensive simulation results show that NarrowGap can effectively accommodate concurrent requests with different resource demands and outperforms three representative schemes in terms of request service rate and network throughput. In this work, we only considered the difference in resource requirements between concurrent requests. We will design a complete resource allocation scheme to provide a fidelity guarantee in future work.

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Zhonghui Li (Member, IEEE) received the bachelor's degree from the School of Information and Software Engineering, University of Electronic Science and Technology of China, in 2018, and the Ph.D. degree from the School of Cyber Science and Technology, University of Science and Technology of China (USTC), in 2023. He is currently a Post-Doctoral Researcher with the School of Cyber Science and Technology, USTC. His research interests include quantum networking and network security.



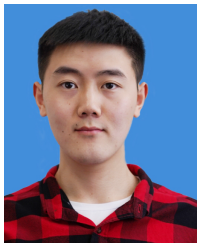
Jian Li (Senior Member, IEEE) received the bachelor's degree from the Department of Electronics and Information Engineering, Anhui University, in 2015, and the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), in 2020. From November 2019 to November 2020, he was a Visiting Scholar with the Department of Electronic and Computer Engineering, University of Florida. From December 2020 to December 2022, he was a Post-Doctoral Researcher

with the School of Cyber Science and Technology, USTC. He is currently an Associate Researcher with the School of Cyber Science and Technology, USTC. His research interests include quantum networking, wireless networks, and next-generation internet architecture. He serves as an Editor for *China Communications*.



Kaiping Xue (Senior Member, IEEE) received the bachelor's degree from the Department of Information Security, University of Science and Technology of China (USTC), in 2003, and the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), USTC, in 2007. From May 2012 to May 2013, he was a Post-Doctoral Researcher with the Department of Electrical and Computer Engineering, University of Florida. Currently, he is a Professor with the School of Cyber Science and Technology, USTC. He is also the

Director of the Network and Information Center, USTC. His research interests include next-generation internet architecture design, transmission optimization, and network security. He is an IET Fellow. His work won Best Paper Awards in IEEE MSN 2017 and IEEE HotICN 2019, the Best Paper Honorable Mention in ACM CCS 2022, the Best Paper Runner-Up Award in IEEE MASS 2018, and the Best Track Paper in MSN 2020. He serves on the Editorial Board for several journals, including IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY (TIFS), IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING (TDSC), IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS (TWC), and IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT (TNSM). He has served as a Guest Editor for many reputed journals/magazines, including IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS (JSAC), *IEEE Communications Magazine*, and IEEE NETWORK. He has also served as the Track/Symposium Chair for some reputed conferences, including IEEE ICDCS, GLOBECOM, WCSP, ICC, and QCNC.



Lutong Chen (Graduate Student Member, IEEE) received the bachelor's degree from the School of Cyber Science and Technology, University of Science and Technology of China, in 2020. He is currently pursuing the Ph.D. degree with the School of Cyber Science and Technology. His research interests include quantum networking and network security.



Nenghai Yu received the B.S. degree from Nanjing University of Posts and Telecommunications, Nanjing, China, in 1987, the M.E. degree from Tsinghua University, Beijing, China, in 1992, and the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), Hefei, China, in 2004. Currently, he is a Professor with the School of Cyber Science and Technology and the School of Information Science and Technology, USTC. He is the Executive Dean of the School of

Cyber Security, USTC, and the Director of the Information Processing Center, USTC. He has authored or co-authored more than 130 papers in journals and international conferences. His research interests include multimedia security, information hiding, and quantum networking.



Qibin Sun (Fellow, IEEE) received the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), in 1997. He is currently a Professor with the School of Cyber Science and Technology, USTC. He has published more than 120 papers in international journals and conferences. His research interests include multimedia security and network intelligence and security.



Jun Lu received the bachelor's degree from Southeast University in 1985 and the master's degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), in 1988. Currently, he is a Professor with the Department of EEIS, USTC. He is also the President with Jiaxing University. He is an Academician with the Chinese Academy of Engineering (CAE). His research interests include theoretical research and system development in the field of integrated electronic

information systems, networks, and information security.